

in Figure 1a indicate the expectation a random vector will lie in the tangent space. The expectation is approximately $\sqrt{n/d}$, where n and d are the dimensions of the tangent space approximation and manifold, respectively [9]. An explanation is therefore sufficiently aligned with the tangent space when that fraction in the tangent space is greater than $\sqrt{n/d}$. We see in Figure 1a that standard base-point choices on CIFAR10 are significantly below the vertical line. It is the goal of future work to determine if the dimension of the tangent space of CIFAR10 of $n = 144$ or the parameter of the Gaussian base-point impacts the tangential alignment of IG on CIFAR10.

We provide in Figure 2, example integrated gradient explanations for a point on MNIST32, FER2013, and Fashion-MNIST with differing base-point choice. We see that our method provides tangentially aligned explanations with $\mu_x > 0.91$ for all datasets. The tangentially aligned integrated gradient attributions are clear and perceptually aligned with the object to classify in the image. We see in Figure 2 that uniform, maximum ℓ_2 distance, and Gaussian are consistently random noise.

4.5. Comparison of Gradient Explainability Models with Tangential Integrated Gradients

In this section we compare tangentially aligned integrated gradients with three common gradient explainability models: Gradient, Smooth Grad and Input * Gradient. The aforementioned gradient explainability models do not require a base-point choice. We demonstrate that tangentially aligned integrated gradients significantly improves upon integrated gradients. The gradient explainability models for a given model are defined as follows:

1. **Gradient** The gradient of a model f at $x \in \mathbb{R}^d$ for class i is defined as:

$$\text{grad}(x)_i := \frac{\partial f(x)_i}{\partial x}. \quad (33)$$

2. **Smooth Grad** We define Smooth Grad with n samples and standard deviation σ as:

$$\text{SmoothGrad}(x) = \frac{1}{n} \sum_{i=1}^n \nabla f(x + a), \quad (34)$$

where, $a \sim \mathcal{N}(0, \sigma^2)$. Following [9] we take $\sigma = 0.02$ and $n = 25$.

3. **Input * Gradient** Input * Gradient is defined as:

$$\text{Input} * \text{Gradient} := x \odot \frac{\partial f(x)_i}{\partial x}. \quad (35)$$

In Figure 1b we provide density plots of the fraction an attribution is in the tangent space for: Gradient, Smooth Grad, Input * Gradient and tangentially aligned integrated gradients. We see in Figure 1b that tangentially aligned integrated gradients provides attributions consistently in the tangent space, out-performing the aforementioned gradient explainability models. In Figures 1a and 1b Gradient, Smooth Grad and Input * Gradient provide better tangential alignment than the common base-point choices provided in Section 2.2 on MNIST and CIFAR10. On Fashion-MNIST we see that the zero base-point choice provides comparable performance with Gradient, Smooth Grad and Input * Gradient. In Figures 1a and 1b we see that on FER2013, Gradient, Smooth Grad and Input * Gradient perform similarly to Gaussian, maximum ℓ_2 distance and zero base-point choices for integrated gradients. All gradient models on FER2013 outperform the uniform base-point choice for integrated gradients. We see in Figures 1a and 1b, Gradient, Smooth Grad, and Input * Gradient tend to out-perform Integrated gradients with standard the standard base-point choices. Tangential integrated gradients out-performs the aforementioned gradient explainability models and standard base-point choices.